LAI Sing Kwong, 賴星光

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EDUCATION

Hong Kong University of Science and Technology Sep 2024 - Present
- Engineering with an Extended Major in Artificial Intelligence (COMP+AI) - CGA: 3.63/4.30 Hong Kong
- Admission Scholarships (HKD$42,100)
- Dean's list (2025 Spring)

RESEARCH EXPERIENCE

Quantum-Classical Hybrid Reinforcement Learning for Quantitative Trading | HKUST Feb 2026 - Present
- **Experimental Lead:** Took ownership of the empirical workflow for the quantum-representation RL study, turning the research direction into runnable experiments, comparable baselines, and reproducible result tables.
- **Benchmarking System:** Built and maintained the FinRL-based evaluation pipeline for FPQC-SAC against continuous-control, robust RL, and rule-based trading baselines, including SAC, TD3, PPO, DDPG, A2C, REDQ, and TQC.
- **Ablation, Diagnostics & Release:** Conducted ablation and diagnostic studies on reward design, state representation, and cross-market behavior; prepared the open-source release: ZeyuLIU-UST/FPQC-SAC-main.

Structured Memory and View Routing for Heterogeneous Investor Simulation | HKUST Feb 2026 - Jun 2026
- **Cognitive Architecture Design:** Replaced static persona prompting with an endogenous cognition framework built on structured episodic memory, prototype induction, and auditable view selection to simulate heterogeneous market participants.

An LLM-based Multi-agent System for Financial Reasoning | HKUST Jun 2025 - Nov 2025
- **Model Validation & Benchmarking:** Responsible for reproducing baseline experiments and executing the core backtesting pipeline to validate model performance. The research project received a top reviewer score of 8/10 at ICLR, cited for its strong quantitative evidence and soundness.
- **System Optimization & Performance Improvement:** Identified a critical bottleneck in the memory module and proposed an architectural enhancement that improved backtesting performance: Microsoft (MSFT) returns increased from negative to +18%; Coinbase (COIN) returns increased from negative to +10%.

PROJECT

AStockArena - Real-Time Audit Benchmark for Financial Agents | HKUST (Team Lead) Oct 2025 - Present
- **Research & System Development:** Contributed to the research concept and implemented the complete codebase for a real-time financial-agent benchmark, including ReAct workflows, market-data and news tools, persistent portfolio accounts, execution constraints, and fault-isolated model orchestration.
- **Decision-Process Auditing:** Built an eight-check audit pipeline spanning execution traceability, evidence sufficiency, and financial meta-awareness; aligned model reports, trade requests, execution responses, and account changes to verify actions, evidence use, and portfolio-state assessments against expert consensus labels.
- **Experimental Execution & Analysis:** Ran all experiments, including the main evaluation of 13 LLM agents across 10 A-share securities, 82 trading days, and 3,198 decisions; implemented historical trade-filtering replay, 1,000 random-control repetitions, slippage sensitivity analysis, historical case reruns, and new-model validation. Code release planned.

PUBLICATIONS & WORKING PAPERS

- Zeyu LIU*, **LAI Sing Kwong***, Xuanzhi Feng*, HaoYang Shang*, Yang LI, Zihao Fang, Xuan Liu. "AStockArena: Beyond Returns, Can We Entrust Real Capital to the Decisions of Financial Agents?" Submitted to ACL ARR.
- **LAI Sing Kwong***, Yuanchen GAO*, Zeyu LIU*, Xuanzhi FENG*, Hualei ZHANG, Yang LI, Xiyue LIU, Jie ZHANG, Song GUO. "Mitigating Bias in Low-SNR Financial Reinforcement Learning via Quantum Representations." Preprint. https://arxiv.org/pdf/2606.10448
- Zeyu LIU*, Xuanzhi FENG*, Haoran XU*, **LAI Sing Kwong***, Yang LI, Hualei ZHANG, Hao CHEN, Jie ZHANG, Xiaoyi PANG, Song GUO. "Foresee to Trade: Forecast-Augmented Policy Optimization for Portfolio Management." Submitted to WWW.
- Zeyu LIU, Jie ZHANG, HaoYang Shang, **LAI Sing Kwong**, Yang LI, Song Guo. "FinThink: An LLM-based Multi-agent System for Financial Reasoning." https://openreview.net/pdf?id=vm7xqrU345

(* Denotes equal contribution)

INTERNSHIP EXPERIENCE

Intelligent Design Technology Limited | ***Digital Marketing Data Analyst*** Jan 2026
- **GenAI-Driven Development:** Engineered a GenAI "Picture Story Generator" for K-6 education, implementing context-aware narratives and optimizing system prompts to ensure safety and age-appropriateness.
- **System Architecture:** Architected modular state-machine workflows for robotic control systems, successfully lowering response latency.

ADDITIONAL INFORMATION

Languages: Native Mandarin, Native Cantonese; Fluent English
Programming Languages: Python, C++